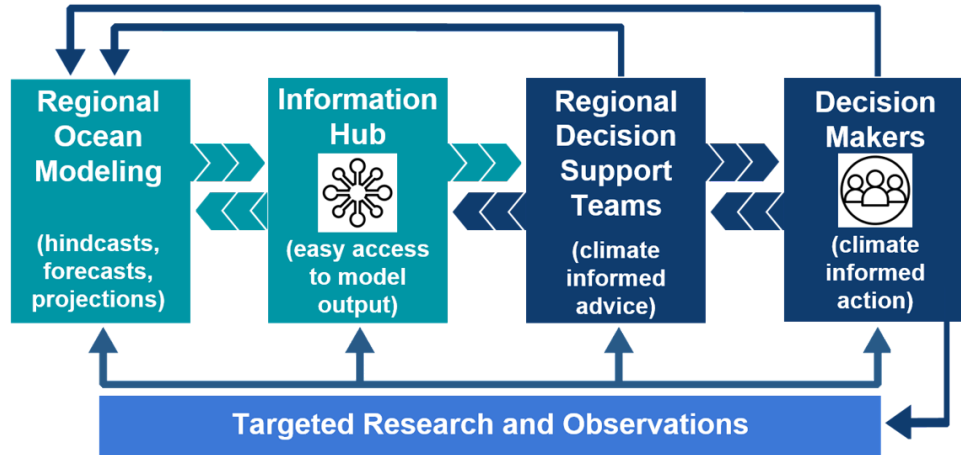


# The Climate, Ecosystems, and Fisheries Initiative (CEFI)

The Climate, Ecosystems, and Fisheries Initiative (CEFI) is a cross-NOAA effort to build the nation-wide, operational ocean modeling and decision support system needed to reduce impacts, increase resilience and help marine resources and resource users adapt to changing ocean conditions.

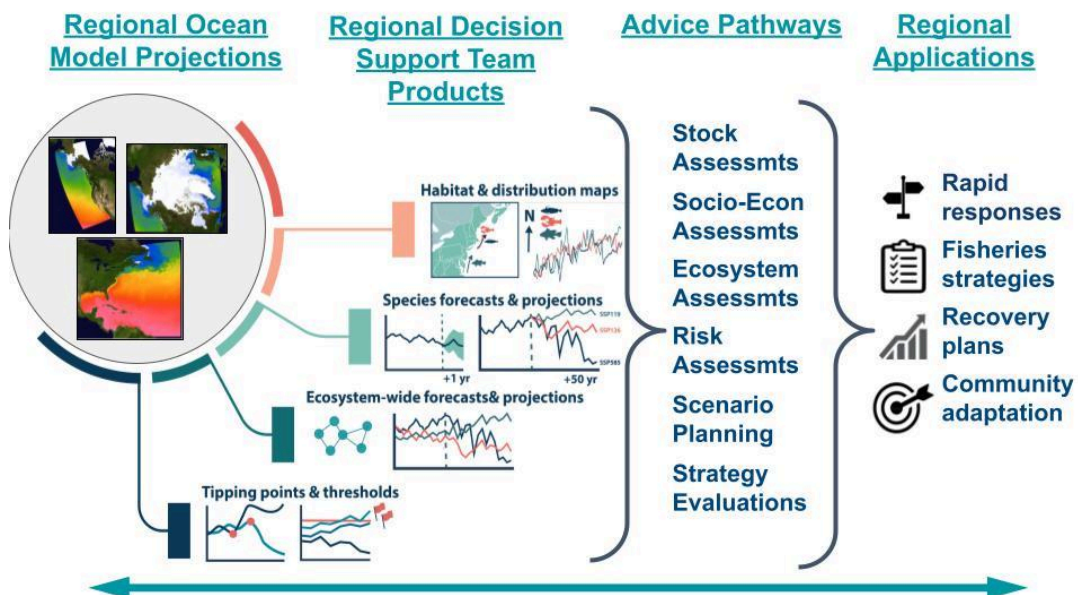
## CEFI Decision Support System



Regional Decision Support Teams will provide the operational capacity to produce the climate-related information and advice needed by decision makers for effective resource management of fisheries, protected species, and protected areas, industry planning, and community adaptation. The Teams will work as part of NOAA's Regional Fisheries Science Centers to provide:

- Early warnings and projections of future ecosystem conditions, such as future species distributions and abundance
- Risk assessments
- Actionable advice for climate-ready fisheries management, protected species conservation, protected area management, and community adaptation

The CEFISystem will increase the capacity of resource managers, fishing communities, and other decision makers to use climate-related information and advice to take action to reduce impacts and adapt to changing ocean conditions.



# The Climate, Ecosystems, and Fisheries Initiative (CEFI)

## Examples of CEFI products and information considerations:

- Eco-climate hindcast, forecast, and projection maps
- Climate-enhanced stock assessments
- Ecosystem Status Reports
- Risk assessments
- Adaptation assessments
- Climate-informed management strategy evaluation
- Climate-informed essential fish habitat
- Portfolio planning tools
- Protected species hotspots
- Near-real time protected species/bycatch risk maps
- Species risk profiles
- Climate refugia maps
- Tipping point detection and risk
- Climate-informed targets & reference points
- Contingency and emergency planning and response
- Dynamic management tools

## CEFI pilot projects and emerging regions:

### Alaska

- [The Alaska Climate Integrated Modeling Project](#) (CEFI pilot project - Bering Sea)
  - Evaluating impacts of changes in temperature, ocean acidification, and oxygen depletion on habitat quality and species distributions and how these changes may impact food web structure
  - Species-distribution models for commercially important species
  - Modeling effects of temperature on recruitment, productivity
  - Modeling of Northern fur seal age-specific habitat selection and predict current and future habitat suitability
  - Understanding changes in the phenology of prey resources ([Cheng et al. 2021](#))
- [Gulf of Alaska Climate Integrated Modeling Project](#) (CEFI pilot project - Gulf of Alaska)
  - Modeling species distribution for ground fish species of commercial interest (e.g., arrowtooth flounder) and can inform EFH

### Northeast

- [The Northeast Climate Integrated Modeling Project](#) (CEFI pilot project - Northeast)
  - Dynamic Range Models for testing management strategies for fish stocks undergoing distribution shifts
  - Spatio-temporal species distribution models
  - Identifying relevant ecosystem and climate influences on stocks and their link to stock dynamics (e.g., distribution and habitat use, recruitment, mortality)
  - [NCLM GitHub](#)

### Southwest

- [The Future Seas Project](#) (CEFI pilot project - West coast)
  - Produced species distribution models for sardine, anchovy, and swordfish in the California Current ([Smith et al. 2023](#))
  - Integration of species distribution models into Atlantis end-to-end model will allow for assessing ecosystem impacts of shifting distributions

## The Climate, Ecosystems, and Fisheries Initiative (CEFI)

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